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Mouse Controller

Use the Arduino Due as a USB host for a mouse.

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Introduction

The Arduino Due has the ability to act as a USB host for peripherals such as mice connected to the SerialUSB port. This example demonstrates the use of the `MouseController` library.

Goals

Learn how to use the Arduino Due as a USB host for a mouse.

Hardware & Software Needed

- Arduino Due** Board
- USB mouse
- Arduino IDE (**online** or **offline**).
- USBHost** library

The Circuit

There is no circuit for this tutorial.

Programming the Board

1. First, let's make sure we have correct the drivers installed. If we are using the Cloud Editor, we do not need to install anything. If we are using an offline editor, we need to install it manually. This can be done by navigating to **Tools > Board > Board Manager....** Here we need to look for the **Arduino SAM boards (32-bits ARM Cortex-M3)** and install it.

2. Now, we need to install the libraries needed. Simply go to **Tools > Manage libraries...** and search for **USBHost** and install it.

The sketch can be found in the snippet below. Upload the sketch to the board.

Code

Before we begin, let's take a look at some of the core functions in the program:

`USBHost` - USBHost is the base class for all calls that rely on USB host communication. When invoked, it initializes a USB controller.

```
MouseController  
mouse(usb);
```

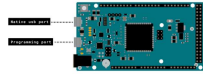
- MouseController is the class for all calls to the USBHost relating to an

`mouseMoved()` - A function that is called whenever a connected USB mouse moves.

```
1  /*
2
3  Mouse Controller Ex
4
5  Shows the output of
6
7  the Native USB port
8
9  created 8 Oct 2012
10
11 by Cristian Maglie
12
13 http://arduino.cc/e
14
15 This sample code i
16
17 */
18
19 // Require mouse con
20 #include <MouseContr
21
22 // Initialize USB Co
23
24 USBHost usb;
25
26 // Attach mouse cont
27
28 MouseController mous
```

Testing It Out

After you have uploaded the code, plug your mouse into the Native USB port on the Due. It enables the Due to emulate a USB mouse or keyboard to an attached computer. In the **Arduino IDE**, open the serial monitor and start clicking and



Arduino Due Ports

Troubleshoot

If the code is not working, there are some common issues we can troubleshoot:

You are using the incorrect USB port

You have not installed the correct drivers for the Arduino Due.

You have not installed the [USBHost library](#).

Conclusion

The Arduino Due has a number of facilities for communicating with a computer, another Arduino or other microcontrollers, and different devices like phones, tablets, cameras and so on. In this example, we have learned how to use the Arduino Due as a USB host for a mouse.

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